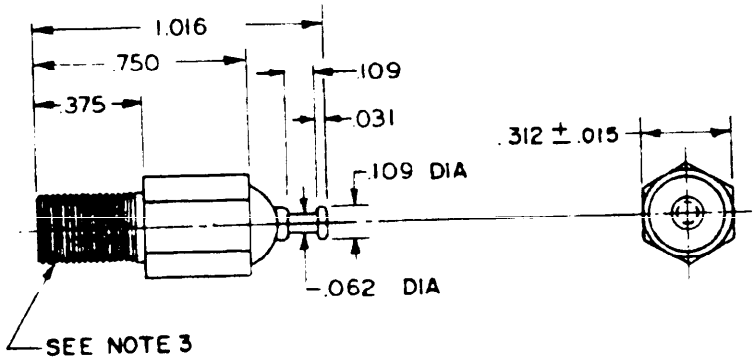
MIL-C-11015/14C
6 DECEMBER 1967
SUPERSEDING
MIL-C-11015/14B
2 April 1962

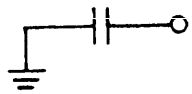
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE) (STAND-OFF), STYLE CK80

The complete requirements for procuring the capacitors described herein shall consist of this document and the issue in effect of Specification MIL-C-11015.

INCHES	MM
.015	.38
.031	.79
.062	1.57
.109	2.77
.312	7.92
.375	9.53
.750	19.05
1.016	25.81





CIRCUIT DIAGRAM

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Threaded portion shall be 1/4-28 UNF-2A and shall extend to within .047 (1.19 mm) of the shoulder.
4. A nut and internal tooth lockwasher shall be supplied with each capacitor.
5. Unless otherwise specified, tolerance is ±.031 (.79 mm).

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See above figure.
Case type - Stand-off.
Capacitance value - See table I.
Capacitance tolerance - ±10 percent (K) or ±20 percent (M) as shown in table I.
Rated temperature - -55° to +85° C.

Dielectric withstanding voltage:
Dielectric (DWV): Method 301 of MIL-STD-202.
Test voltage - 250 percent of rated voltage, applied between the case and terminal.
Body insulation: Not applicable.

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 20261O-17T14:55Z
Check the source to verify that this is the currentversion before use.

TABLE I. Style CK80 characteristics.

Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	volts, dc		pf	
CK80AX100-	500	AX	10	K, M
CK80AX330-	500	AX	33	K, M
CK80AX470-	500	AX	47	K, M
CK80AX680-	500	AX	68	K, M
CK80AX101-	500	AX	100	K, M
CK80AX471-	500	AX	470	K, M
CK80AX681-	500	AX	680	K, M
CK80AW102M	500	AW	1,000	M
CK80AW152M	500	AW	1,500	M

1/ Where applicable, the complete type designation will include an additional symbol to indicate capacitance tolerance.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition B (50,000 ft).
Test potential - 150 percent of rated voltage, applied between the case and terminal.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 20,000 megohms, min.

Capacitance (Cap.): Method 305 of MIL-STD-202.
Test frequency:
100 pf or less - 1 MHz ± 1 kHz.
Over 100 pf - 1 kHz ± 100 Hz.
Test voltage - 1.0 ± 0.2 volt rms.

Dissipation factor (DF): 1.5 percent, max (voltage-temp limit X) or 2.0 percent, max (voltage-temp limit W).

Shock, medium impact: Method 205 of MIL-STD-202, condition C (50 G's).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G's).

Temperature cycling and immersion:
Temperature cycling: Method 102 of MIL-STD-202, condition D.
Immersion: Method 104 of MIL-STD-202, condition B.
IR - 3,000 megohms, min.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Terminal strength: In accordance with MIL-C-11015, 5 pounds applied.

Moisture resistance: Method 106 of MIL-STD-202.
DWV - 250 percent of rated voltage.
IR - 3,000 megohms, min.
Cap. - Within tolerance of table I value.

Solderability: Method 208 of MIL-STD-202.
1 terminal.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
IR - 20,000 megohms, min.
ΔCap. - +5 percent of initial measurement.
ΔDF - 0.5 percent, max.

Voltage-temperature limits: In accordance with MIL-C-11015.

Life (at elevated ambient temperature):
Method 108 of MIL-STD-202. 2,000 hours (for qualification); 250 hours (performance check); 1,750 hours (continuation test).
Test potential - 200 percent of rated voltage.
IR - 500 megohms, min (at 85° C) and 10,000 megohms, min (at 25° C).
DF - 3.0 percent, max (voltage-temp limit X) or 4.0 percent, max (voltage-temp limit W) (at 25° C).

Marking: In accordance with MIL-C-11015.

Source: https://assist.dla.mil -- Downloaded: 202610-17T14:55Z
Check the source to verify that this is the current version before use.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MI
Navy - EC, AS, OS, SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

(Project 5910-0829-12)

Code "C"

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:55Z
Check the source to verify that this is the current version before use.